TRACELOCK – A SECURE DOCUMENT SHARING & TASK MANAGEMENT SYSTEM



Presented By -

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Problem Statements

The Problems

- Files can be shared without any control or restrictions, exposing sensitive data to unauthorized users.
- No tracking of file access — organizations cannot tell who viewed or downloaded a document.
- No proper task assignment system to delegate work and monitor employee progress effectively.
- Organizations credential data can be get leaked.

Introduction

Secure File Sharing Platform

This project is a secure platform for sharing files between Admin and Employees, ensuring only authorized users can access sensitive documents.

Integrated Task Management

It combines file sharing with task management features, allowing admins to assign work alongside the documents employees need to complete it.

Data Security First

The system ensures data security using watermarking and controlled access, preventing unauthorized distribution of sensitive files.

Organizational Efficiency

It helps organizations manage documents and employee tasks efficiently, improving productivity and accountability across teams.

Technologies Used

The system is built on a modern, robust technology stack designed for performance, security, and scalability.



Frontend

React.js — A powerful JavaScript library for building dynamic, responsive user interfaces.



Backend

Node.js & Express.js — A fast, scalable server-side runtime and web framework for handling API requests.



Database

MySQL — A reliable relational database for storing users, files, tasks, and submission records.



Authentication

JWT (JSON Web Tokens) — Stateless, secure token-based authentication for all user sessions.



File Handling

Multer — A Node.js middleware for handling multipart form data and file uploads efficiently.



PDF Processing

pdf-lib — A JavaScript library for creating and modifying PDFs, used for applying watermarks.

System Features

User Authentication

Separate Admin and Employee roles with secure JWT-based login and access control.

File Upload & Sharing

Admins can upload documents and share them directly with specific employees or groups.

Group Management

Organize employees into groups for streamlined file distribution and task assignment.

Notification System

Employees receive instant notifications when new files or tasks are assigned to them.

Task Assignment

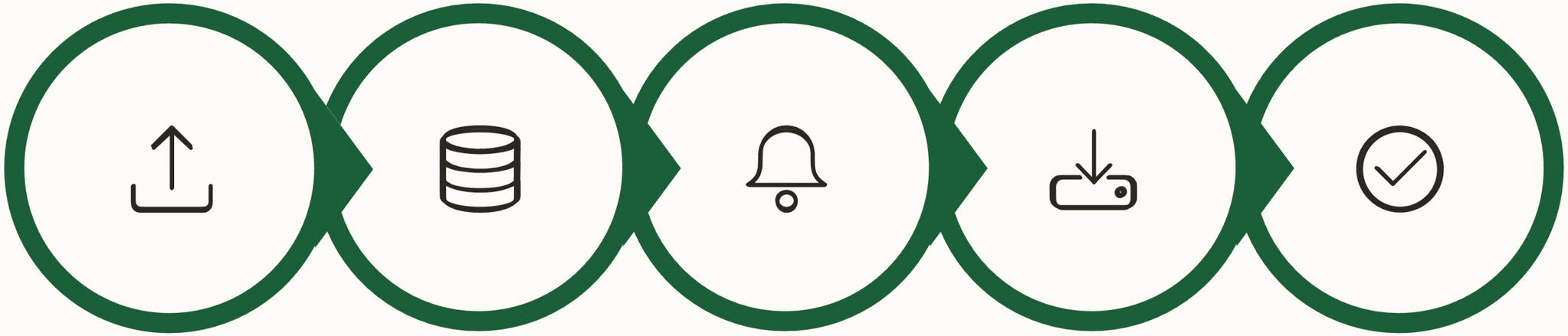
Admins can create tasks with clear deadlines and link them to relevant shared documents.

File Watermarking

Every downloaded file is automatically stamped with the Employee ID and download date to prevent leakage.

System Working Flow

The end-to-end workflow connects admin actions to employee deliverables through a secure, trackable pipeline.



This streamlined flow ensures every document interaction is logged, every task is traceable, and every submission is captured — giving administrators complete visibility over the entire document lifecycle.

Conclusion & Future Scope

Conclusion

Secure File Sharing & Task Management

The system provides a unified, secure platform for sharing documents and managing employee tasks within an organization.

Prevents Data Leakage

Watermarking with Employee ID and date ensures every downloaded file is traceable, significantly reducing unauthorized sharing.

Improves Tracking & Productivity

Deadline-based task assignment and submission tracking improve organizational accountability and overall team productivity.

Future Scope

Advanced Encryption

Add end-to-end encryption for files at rest and in transit for maximum security.

Mobile App Integration

Develop native iOS and Android apps for on-the-go access and task management.

Activity Analytics

Build a dashboard with detailed usage reports, file access logs, and productivity insights.

Thank You !!